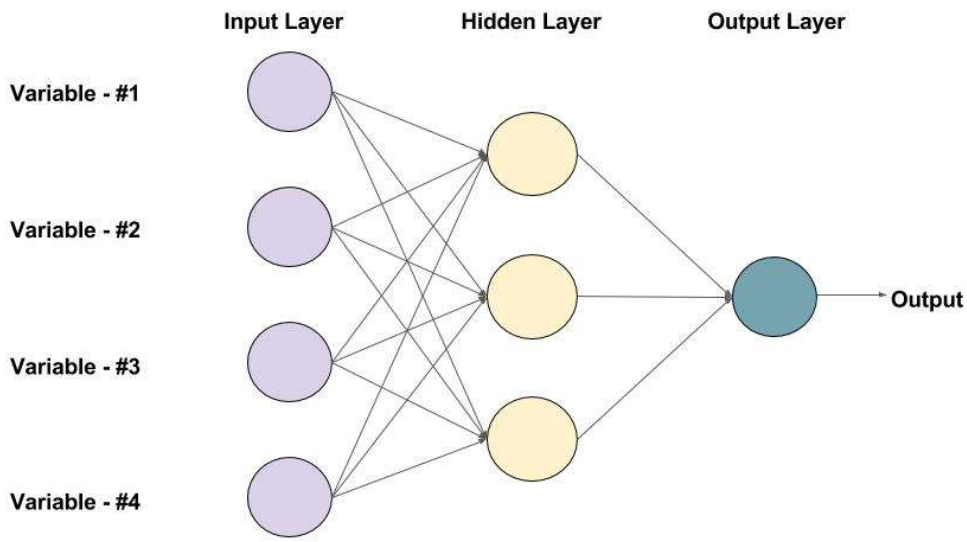
Exercise: 2

Consider the neural network as given below:



An example of a Feed-forward Neural Network with one hidden layer (with 3 neurons)

The activation function of the hidden neurons is:

$$f(z_h) = \frac{1}{1 + e^{-z_h}}$$

Such that:

$$z_h = \sum_{i=1}^{d} w_{ij} x_i \quad \text{and } h = \{1,2,3\}$$

The activation of the output layer is:

$$g(z_o) = z_o = \sum_{j=1}^{3} w_{jk} f(z_j)$$

d is the number of features and x_i are the features.

w_{ij} is the weight between input node i in the input layer and the hidden node j in the hidden layer.

w_{jk} is the weight between hidden node j in the hidden layer and the output node k in the output layer.

The input layer contains the vector of features.

The loss function of the network is computed as:

$$L_S = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

Consider the NASA data set, obtained from a series of aerodynamic and acoustic tests of two and three-dimensional airfoil blade sections conducted in an anechoic wind tunnel.

Use the following URL to access the data:

https://archive.ics.uci.edu/ml/datasets/Airfoil+Self-NoiseTapez une équation ici.

Attribute Information:

This problem has the following inputs:
1. Frequency, in Hertzs.
2. Angle of attack, in degrees.
3. Chord length, in meters.
4. Free-stream velocity, in meters per second.
5. Suction side displacement thickness, in meters.

The only output is:
6. Scaled sound pressure level, in decibels.

Directions:

1- Implement the architecture of this network.
2- Partition your data into 70% of training data and 30% of testing data.
3- Use the given training data to compute the estimated outputs $\hat{y}_i$ and then the overall training loss L_S.
4- Look for the backpropagation technique and give its algorithm.
5- Implement the backpropagation algorithm on this network to propagate the error backward in the network.
6- Update the weights using the gradient descent algorithm.
7- Compute the test error.
8- Compare the training and the testing error, then conclude in terms of overfitting/underfitting.
9- If you notice that the network is not performant, what are you suggested solutions to improve its performance.